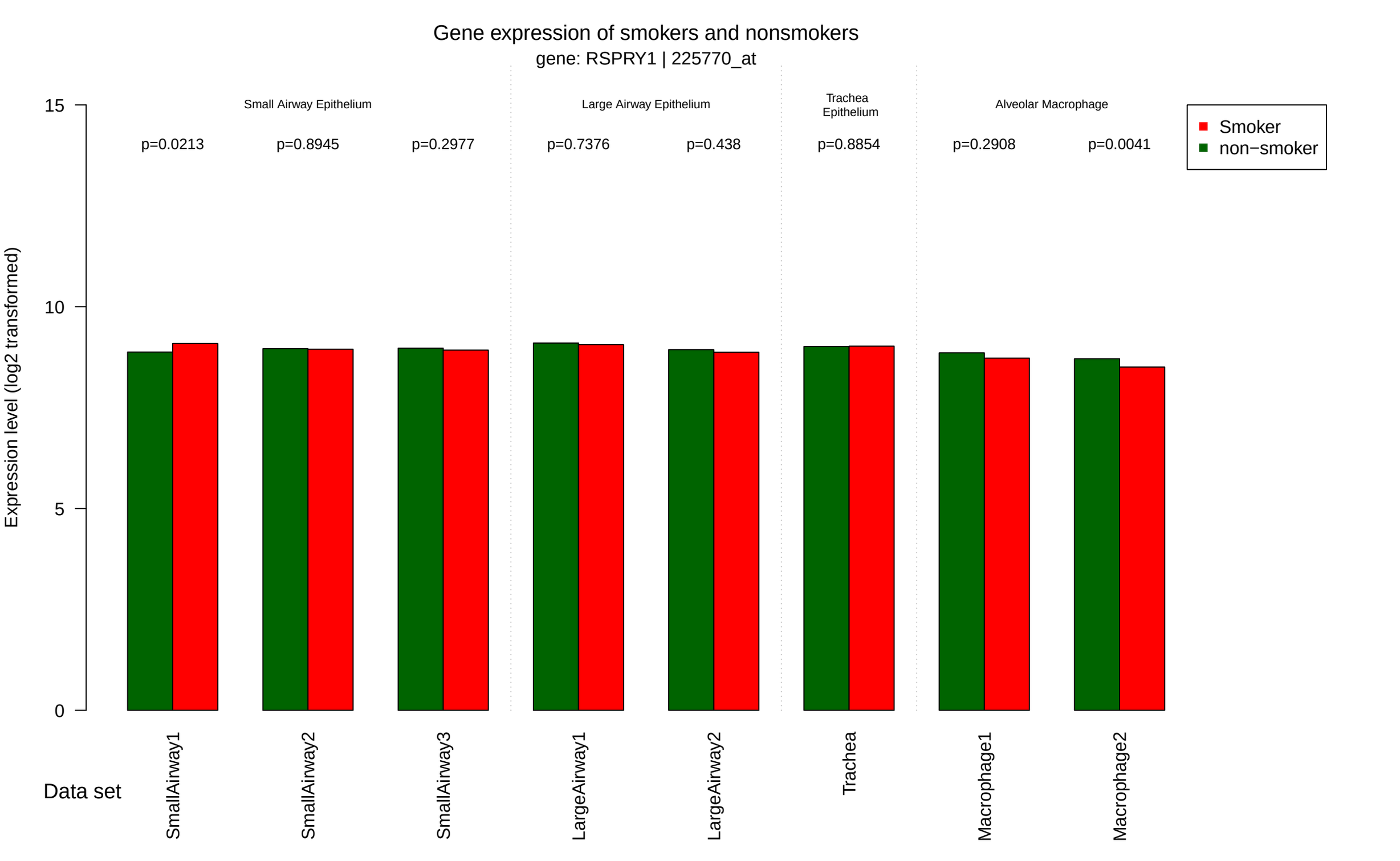
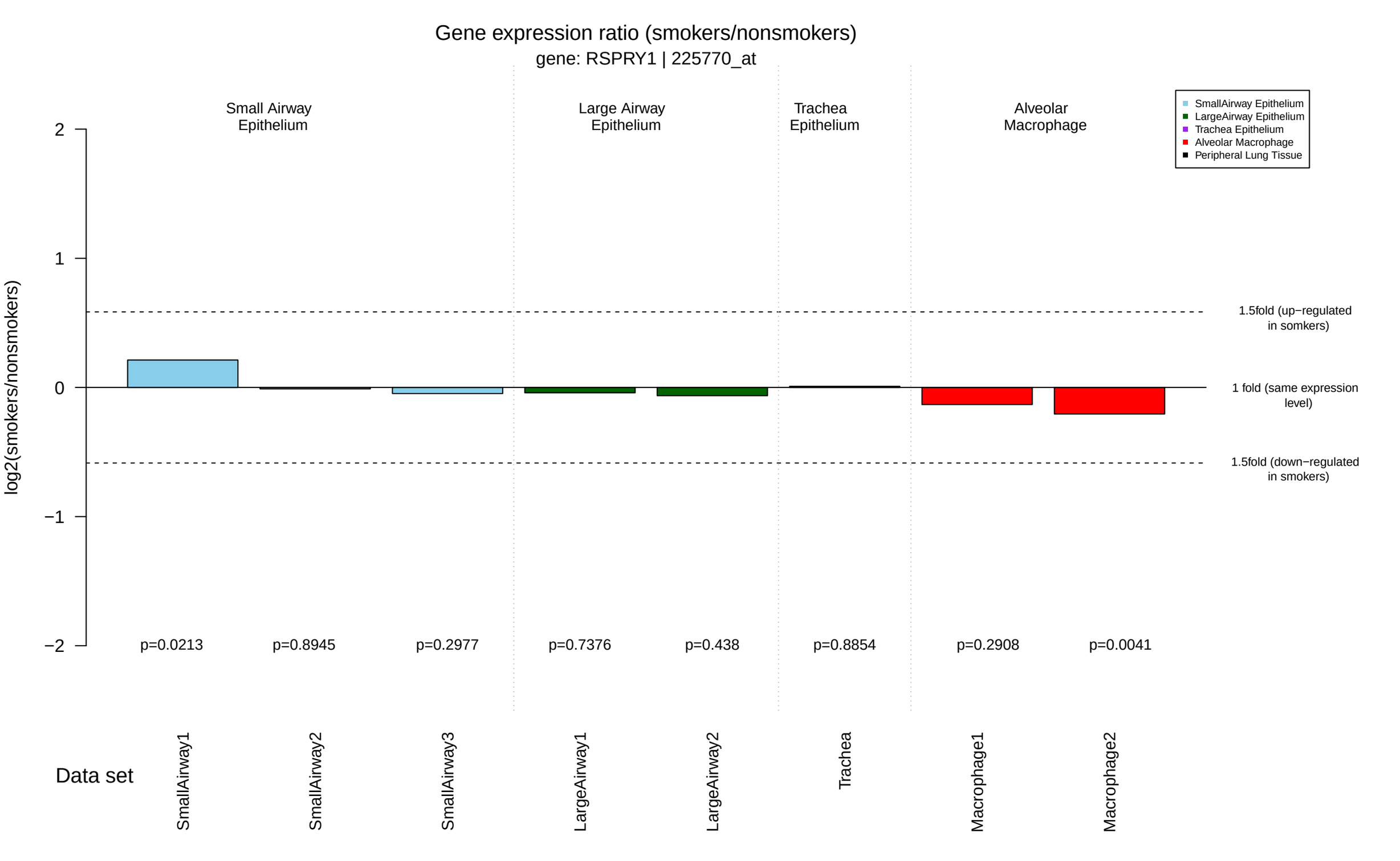
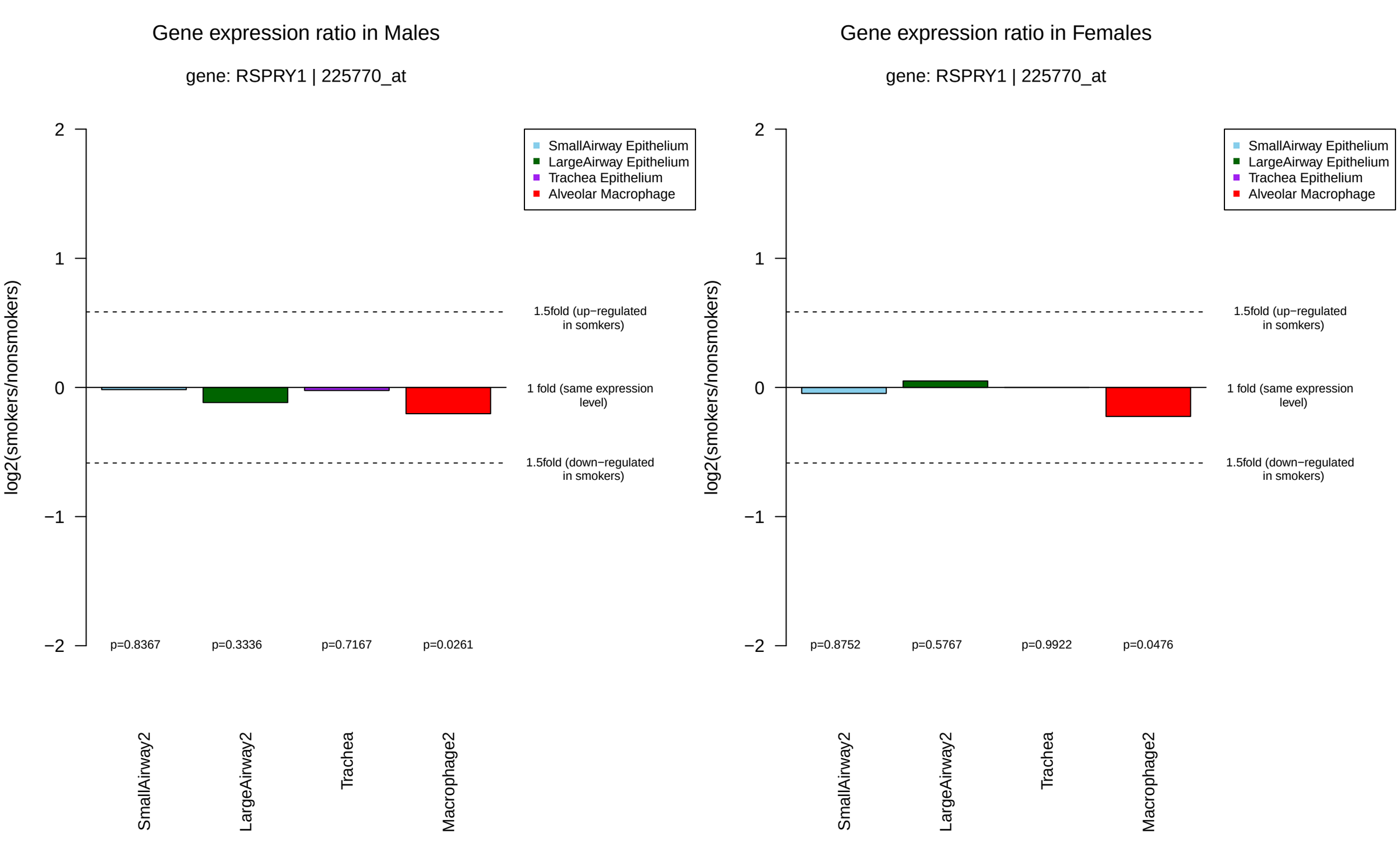
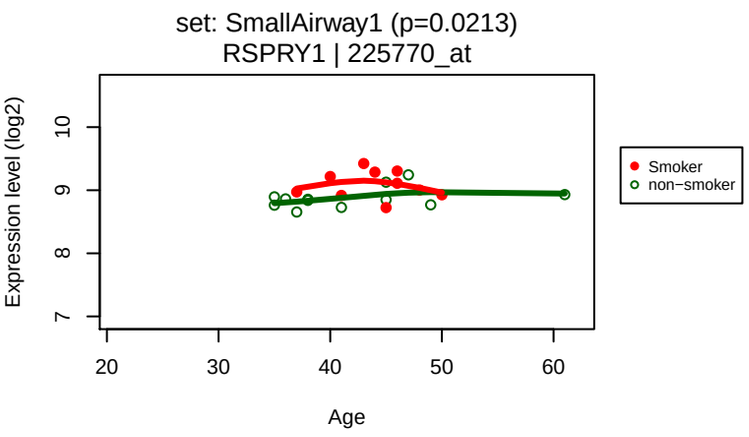
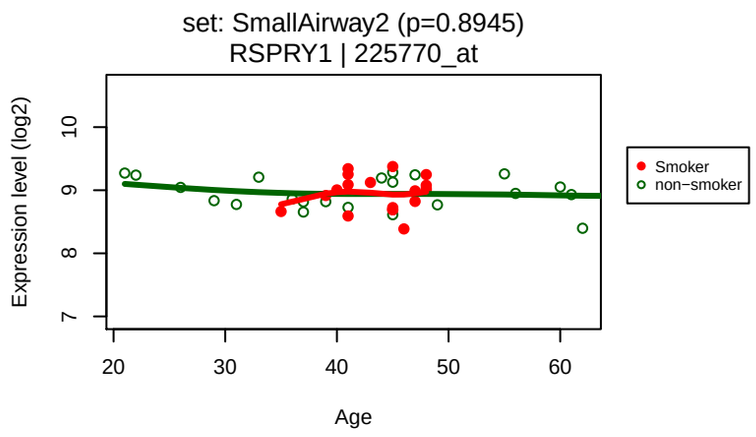
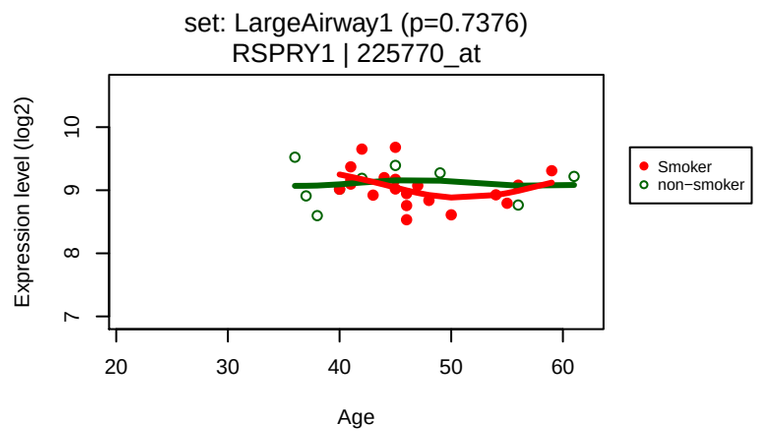

Gene expression of smokers and nonsmokers
gene: RSPRY1 | 225770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0213
p=0.8945
p=0.2977
p=0.7376
p=0.438
p=0.8854
p=0.2908
p=0.0041
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

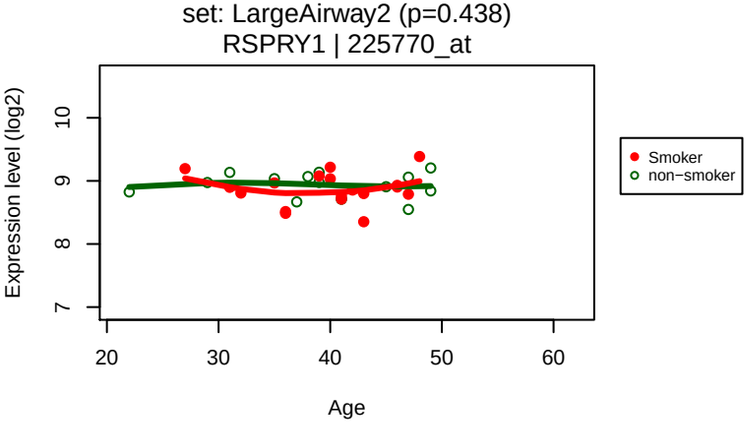

Gene expression ratio (smokers/nonsmokers)
gene: RSPRY1 | 225770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0213
p=0.8945
p=0.2977
p=0.7376
p=0.438
p=0.8854
p=0.2908
p=0.0041
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

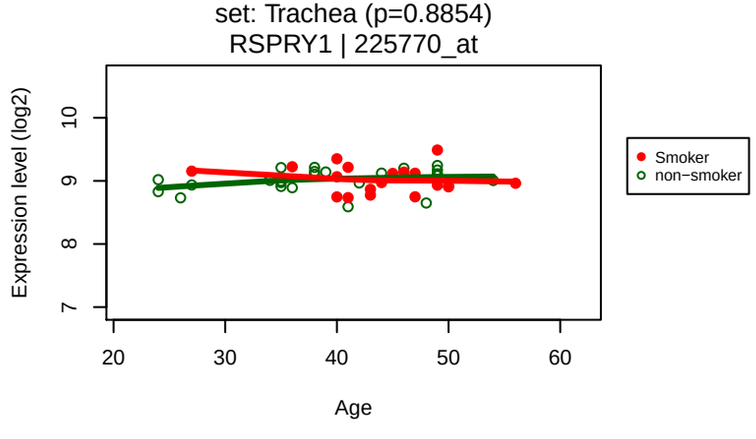

Gene expression ratio in Males
gene: RSPRY1 | 225770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8367
p=0.3336
p=0.7167
p=0.0261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPRY1 | 225770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8752
p=0.5767
p=0.9922
p=0.0476
SmallAirway2
LargeAirway2
Trachea
Macrophage2

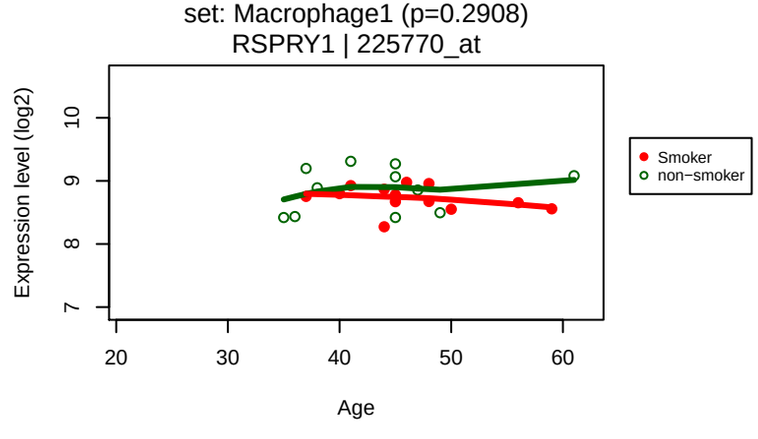

set: SmallAirway1 (p=0.0213)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker

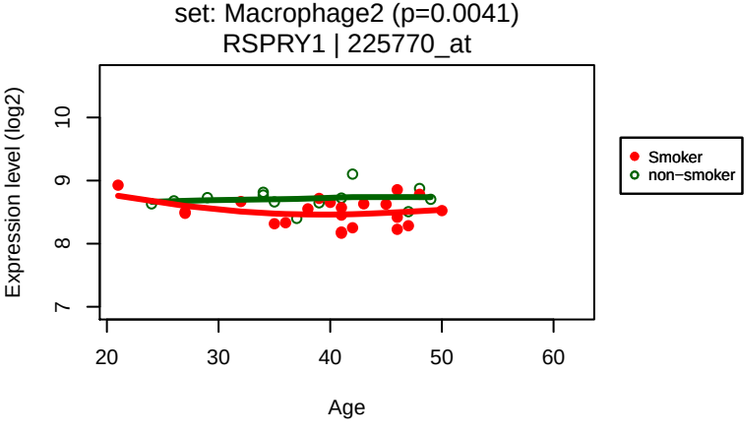

set: SmallAirway2 (p=0.8945)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker

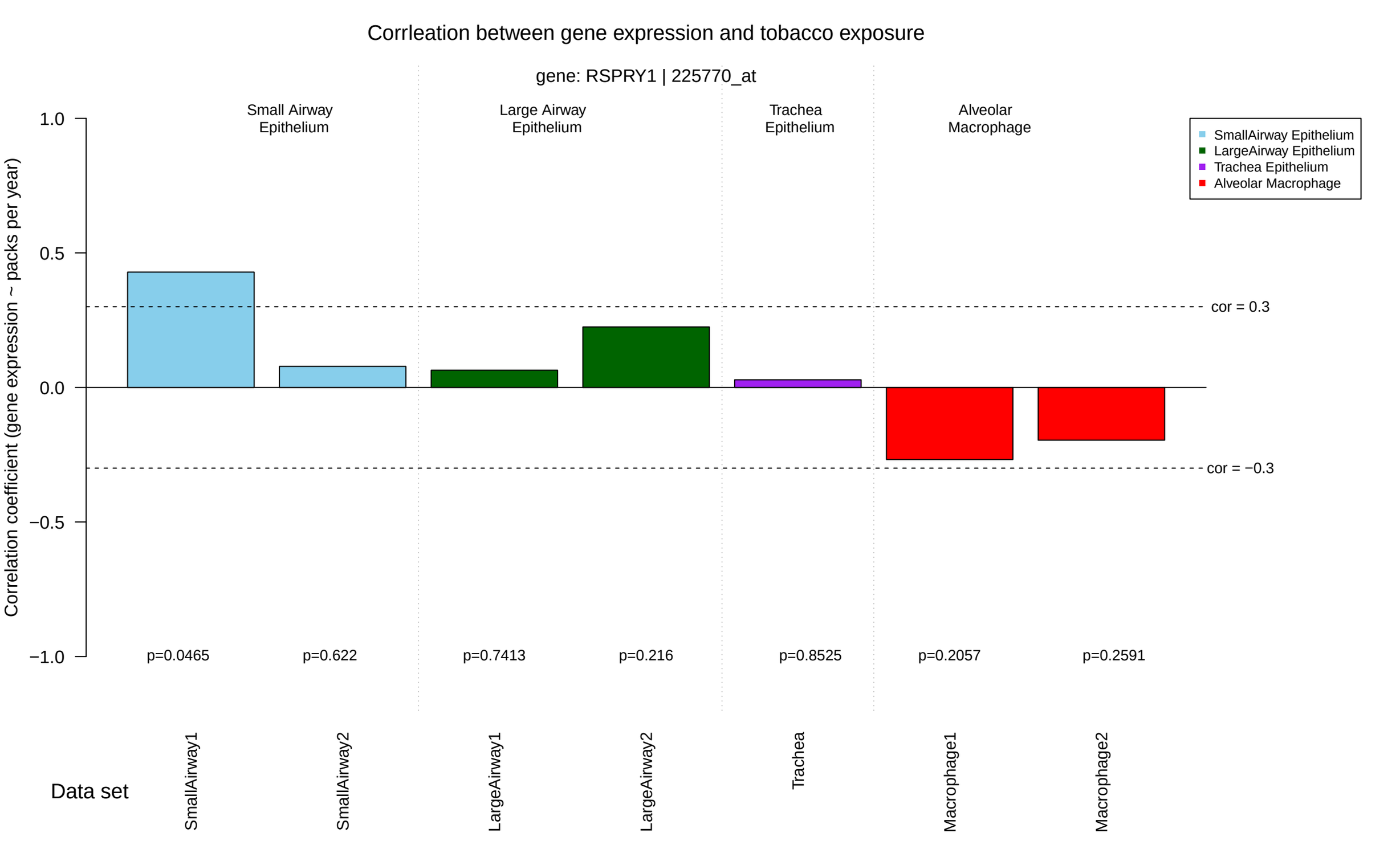

set: LargeAirway1 (p=0.7376)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.438)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8854)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2908)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0041)
RSPRY1 | 225770_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPRY1 | 225770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0465
p=0.622
p=0.7413
p=0.216
p=0.8525
p=0.2057
p=0.2591
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2